Problem 1

(a)Calculate the pairwise correlation matrix.

	x1	x2	x3	x4	x5
x1	1	0.332836	0.418462	0.255285	0.732272
x2	0.332836	1	0.251035	0.287709	0.787538
x3	0.418462	0.251035	1	0.48358	0.684257
x4	0.255285	0.287709	0.48358	1	0.782632
x5	0.732272	0.787538	0.684257	0.782632	1

(b)Calculate the variance of each series, using every observation that series has
(b) variance per column
x1 0.000406074
x2 0.000480066
x3 0.00015111
x4 0.000841739
x5 0.00027315

(c)Is the correlation matrix positive semi-definite? Show how you know.
The answer is False, the correlation matrix is not PSD. This is because the smallest eigenvalue of the correlation matrix is negative. This could lead to issues in risk analysis as it may produce negative variance or other nonsensical results.

Eigenvalues of the correlation matrix: [3.08541647 -0.08927867 0.4613398 0.71908517 0.82343722]

(d) If it is not, use Higham's method to find the nearest positive semi-definite correlation matrix. Report the matrix and its eigenvalues. Which entries changed the most?
Nearest psd correlation matrix:

```
(d) Nearest positive semi-definite correlation matrix:
[[1.         0.3494522  0.42594192 0.27078221 0.69457247]
 [0.3494522  1.         0.26059877 0.30752328 0.73933705]
 [0.42594192 0.26059877 1.         0.4925001  0.66255848]
 [0.27078221 0.30752328 0.4925001  1.         0.73767705]
 [0.69457247 0.73933705 0.66255848 0.73767705 1.        ]]
```

Eigenvalues of the nearest positive semi-definite correlation matrix:

[3.03973389e+00 -6.03526907e-11 4.50081533e-01 7.03476106e-01 8.06708468e-01]

To compare the two matrices, I subtract the corrected psd matrix by the previous pairwise correlation matrix, and below is the resulted difference matrix:

	x1	x2	x3	x4	x5
x1	0	0.0166163	0.00748014	0.0154974	0.0376999
x2	0.0166163	0	0.00956371	0.0198141	0.0482011
x3	0.00748014	0.00956371	0	0.0089197	0.0216987
x4	0.0154974	0.0198141	0.0089197	0	0.0449554
x5	0.0376999	0.0482011	0.0216987	0.0449554	0

We can see that the entry that changes the most is (x2, x5), with a difference of 0.0482011

(e) Combine your results into a covariance matrix. Report it

(e) Repaired covariance matrix:

	x1	x2	x3	x4	x5
x1	0.000406074	0.000154291	0.000105511	0.000158311	0.000231324
x2	0.000154291	0.000480066	7.01891e-05	0.000195487	0.000267728
x3	0.000105511	7.01891e-05	0.00015111	0.000175647	0.000134608
x4	0.000158311	0.000195487	0.000175647	0.000841739	0.000353716
x5	0.000231324	0.000267728	0.000134608	0.000353716	0.00027315

Problem 2

	5%	1%
VaR delta normal %	3.2245	4.5605
ES delta normal %	4.0436	5.2248
VaR historical %	3.0795	4.0731
ES historical %	3.7986	4.5158
VaR simulated %	3.2229	4.5296
ES simulated %	4.0372	5.2082

(a)What is your VaR at the 95% and 99% levels, in percent? Explain your method

I fit the XYZ to the normal distribution of SPY. The closed form equation of XYZ is r_XYZ ~ N(alpha + beta*mu_spy, beta^2*sd_spy^2 + sigma_e^2). This corresponds to the estimated mean and standard deviation of R_XYZ. With that in mind, I plug in these two variables into the pre-written python function for calculating VaR and ES. I used all three methods for this calculation, namely the delta normal method, the monte carlo method and the historical method.

Method/CI	5%	1%
VaR Delta Normal %	3.2245%	4.5605%
VaR historical %	3.0795%	4.0731%
VaR simulated %	3.2229%	4.5296%

Value at risk is the maximum amount of loss a portfolio is likely to encounter during a given time period over the specified confidence interval.

The delta normal method assumes that the returns of XYZ can be approximated by a normal distribution with mean mu_x and standard deviation sd_x, which is derived from the linear regression of XYZ on SPY.

The historical method uses the empirical distribution of past returns of XYZ to calculate VaR

For monte carlo simulation, I first calculated the current position of XYZ based on the last latest stock price, 74.5183, times the shares (100). Which gives me PV= 7451.83. I set the number of trials to be 100,000. It generates a large number of potential future returns for XYZ based on the estimated regression model and the distribution assumptions for SPY and the residuals. And by calculating each simulated return, I empirically estimate the VaR.

(b)What is your ES at the 95% and 99% levels, in percent? Explain your method.

Method/CI	5%	1%
ES Delta Normal %	4.0436%	5.2248%
ES historical %	3.7986%	4.5158%
ES simulated %	4.0372%	5.2082%

Expected Shortfall (ES) is the average of losses that exceed the Value at Risk (VaR) at a given confidence level.

For a normal distribution, ES can be calculated analytically using the standard deviation, mean, and the confidence level.

For historical data, ES is calculated as the average of the worst losses corresponding to the confidence level.

For monte carlo simulation, we first sort the simulated values in ascending order, then take the average of the worst fraction a% of the losses

Problem 3

(a)For each stock, fit a normal, a Student's t, and a normal inverse Gaussian distribution to its returns. Report the fitted parameters. Which distribution fits each stock best? Justify your choice.

```
--- problem 3(a) ---
          Normal             Student t                            NIG                                        Shape
              mu      sigma          nu          mu      sigma   alpha     beta     delta          mu       skew excess kurt
Stock
S1    0.000903697  0.0118202 3.55369e+08  0.00090366  0.0118203 2086.49 -588.204  0.257473    0.076557 -0.0396549  -0.0325185
S2    0.000800265  0.0153399     4.85788 0.000581392  0.0116726 73.2066  4.86382 0.0165931 -0.00030457  0.0705999      5.4772
S3    0.000423612  0.0188885     4.16048   0.0022911  0.0136572 79.8367 -36.3446 0.0196568   0.0104739   -1.32084     4.69642
S4    0.000779986 0.00992765 3.39036e+08 0.000779999 0.00992769 3368.19 -984.958  0.290287   0.0895491 -0.0346262   -0.132614
S5    0.000708771  0.0189416     6.28925 0.000833264  0.0155766 73.5768 -2.23256 0.0259584  0.00149685  -0.307478     2.73261
S6    0.000804694  0.0150776     3.82376  0.00244489  0.0106549 91.6825 -41.6862 0.0146275    0.008272   -1.20212     4.69963
S7    0.000692728  0.0136667      319103 0.000692417  0.0136681 1635.96 -443.713  0.272611   0.0774819 -0.0405896   -0.123222
S8    0.000373999  0.0171031     3.48329 0.000654108  0.0114239 50.6228 -2.88682 0.0141613  0.00118304  -0.443967      6.5288
S9     0.00174602  0.0206364     4.38521  0.00338341  0.0153101 63.1595 -22.9267 0.0215149   0.0101276   -1.08323     4.46136
S10   0.000164798  0.0130547     4.12127 0.000404718 0.00942699 76.6857  -5.3684 0.0125953  0.00104879  -0.696389     5.58253
```

Normal: S1, S4, S7. Fitted t has an enormous nu(df), which signals thin tails. The NIG has a huge α, and the alpha, beta, sigma and mu values are arbitrary. Also, they have low kurtosis.

NIG: S3, S6, S9. They have a large negative beta, which captures their negative skew.

Student t: S2, S5, S8, S10. Their t df is roughly near 5. This means they have fatter tails than normal.

(b)Using the distributions you chose in part (a), fit a Gaussian copula and a Student's t copula. Which one fits better? Justify your choice.

```
          loglik      nu  k     AICc      BIC  dBIC (t vs Gauss)
copula
gaussian 2110.03     NaN  0 -4220.05 -4220.05                NaN
t        2108.16 99.9584  1 -4214.32 -4209.42             -10.635
fitted t-copula nu: 99.9584
min eigenvalue of R: 0.3820457546143446
```

Gaussian fits better. Because the dBIC is -10.635. This means that the BIC of Gaussian (-4220.05) is smaller than t (-4209.42), which indicates that gaussian copula fits the data better.

```
R (Kendall):
         S1     S2     S3     S4     S5     S6     S7     S8     S9    S10
S1        1  0.543 0.5282 0.5618  0.542 0.2941 0.2487 0.2315 0.2811  0.293
S2    0.543      1 0.5637 0.5301 0.5232 0.3296 0.2331 0.2461 0.3104 0.2762
S3   0.5282 0.5637      1  0.532 0.5706 0.3163 0.3026 0.2678   0.34 0.2983
S4   0.5618 0.5301  0.532      1 0.5682  0.306 0.2328 0.2686 0.2962 0.2991
S5    0.542 0.5232 0.5706 0.5682      1 0.3331 0.2699 0.2794 0.2763 0.2959
S6   0.2941 0.3296 0.3163  0.306 0.3331      1 0.5346 0.5244 0.5391 0.5467
S7   0.2487 0.2331 0.3026 0.2328 0.2699 0.5346      1 0.5258 0.5223 0.5421
S8   0.2315 0.2461 0.2678 0.2686 0.2794 0.5244 0.5258      1 0.5204  0.524
S9   0.2811 0.3104   0.34 0.2962 0.2763 0.5391 0.5223 0.5204      1 0.5217
S10   0.293 0.2762 0.2983 0.2991 0.2959 0.5467 0.5421  0.524 0.5217      1
```

(c)Using the copula you chose in part (b), simulate the portfolio with at least 5,000 draws. Calculate the absolute VaR and ES at the 95% level, in dollars, for each stock, each book, and the total portfolio. Present the results in a table. Discuss the results.

	Stock	Book	Holding	price	value	weight
0	S1	A	500	96.659	48329.5	0.110874
1	S2	A	200	233.487	46697.4	0.10713
2	S3	A	300	80.8909	24267.3	0.0556721
3	S4	A	100	435.968	43596.8	0.100016
4	S5	A	800	47.1593	37727.5	0.0865514
5	S6	B	150	308.844	46326.6	0.106279
6	S7	B	250	160.577	40144.3	0.092096
7	S8	B	600	43.4098	26045.9	0.0597525
8	S9	B	300	327.496	98248.7	0.225395
9	S10	B	75	326.832	24512.4	0.0562344

	Book	Value $	VaR 5%	ES 5%	VaR % of value	ES % of value
S1	A	48,329.48	896.32	1,135.86	1.85	2.35
S2	A	46,697.43	1,078.55	1,579.15	2.31	3.38
S3	A	24,267.28	798.96	1,192.94	3.29	4.92
S4	A	43,596.81	677.28	857.47	1.55	1.97
S5	A	37,727.45	1,105.03	1,528.37	2.93	4.05
S6	B	46,326.64	1,198.38	1,853.73	2.59	4.00
S7	B	40,144.31	875.17	1,105.80	2.18	2.75
S8	B	26,045.88	641.44	1,011.49	2.46	3.88
S9	B	98,248.71	3,375.49	5,090.62	3.44	5.18
S10	B	24,512.38	480.42	718.96	1.96	2.93
A		200,618.44	3,566.83	4,820.04	1.78	2.40
B		235,277.92	5,378.87	7,696.22	2.29	3.27
Total		435,896.36	7,530.79	10,420.48	1.73	2.39
Sum of groups		NaN	8,945.70	12,516.27	NaN	NaN
Diversification		NaN	1,414.91	2,095.78	NaN	NaN

sum of stock VaRs: 11127.05 sum of stock ES: 16074.38

- From the results, we can see that Book A diversifies better than Book B. Book A has lower VaR and lower ES than Book B, possibly because A's value is spread evenly, while B is dominated by S9.
- Both VaR and ES are subadditive in this sample: total ≤ sum of books ≤ sum of stocks. ES is inherently subadditive, VaR is subadditive only in this sample.

Under normality, ES/VaR at 5% ≈ 1.25.

- S1, S4 and S7 are close to 1.25, which aligns with my previous normal distribution prediction.
- S6, S8 and S9 are around 1.5–1.6, indicating fat tails. S9 is the most volatile because of its fat tail indicated by the extremely high ES/VaR ratio.